

.NET Lab: Dependency Injection captive lifetime validation

Duration : 35 Min.

Scenario

A performance analysis identifies memory leaks due to Scoped services registered inside Singleton services (captive dependency).

Task Objectives

Perform the following actions inside the `student_workspace` directory: - Refactor DI registrations in `Program.cs` to resolve captive lifetimes.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Navigate to and click the specific files mentioned in the Task Objectives.
3. Make the necessary code edits in the editor.
4. Press `Ctrl + S` (Windows) or `Cmd + S` (Mac) to save your changes.
5. If you need to run commands, open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.

Validation

Once you have saved your files and are ready to submit, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Local VM Environment active and verified	0 Marks
TC2	Program.cs exists in workspace	5 Marks
TC3	Transient / Scoped services are not injected into Singleton class constructors	5 Marks
TC4	Register target scopes correctly under <code>builder.Services</code>	5 Marks

TC5	No syntax or compilation issues in Program.cs	5 Marks
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Total Score: 20 Marks